

WHAT AN ENCODER MODEL IS

One page. No code. Read it once before Day 1.

An encoder reads a piece of text and hands back numbers that stand for its meaning, and that is the whole of what it does. Generative models like ChatGPT continue text instead, which is a different job. This workshop spends its time on the reading kind.

Words become numbers

A computer cannot compare two sentences directly, so the first move in any text-as-data pipeline turns text into numbers. The crude way counts words, which is what a **bag of words** does, and it throws away order and context. Under that scheme **the tax cut helped workers** and **workers cut the tax help** look identical.

An encoder reads the whole sentence at once and gives every word a set of numbers shaped by the words around it. So **bank** in a sentence about rivers comes out different from **bank** in a sentence about lending. Those numbers are called **embeddings**.

Why the name

The family is called BERT, short for Bidirectional Encoder Representations from Transformers. **Bidirectional** means it reads left to right and right to left together. **Encoder** means it compresses text into that numeric representation. You will meet BERT, RoBERTa and DeBERTa, all variants of the same idea, plus DistilBERT, a smaller and faster version.

What we do with it

An encoder on its own produces only numbers. To get a label you bolt a small classifier on top, show it examples, and let it learn the mapping from numbers to labels. That is **fine-tuning**, and it is the Day 3 capstone. A model trained on a few hundred hand-coded sentences can then code hundreds of thousands, which is the reason to bother.

Three words to carry in

Token

A chunk of text the model reads, usually a word or part of one.

Embedding

The list of numbers standing for a token or a sentence.

Fine-tuning

Training a pretrained model further on your own labeled examples.